Year 9 5.2 Yearly Exam Revision

Help sheet and answer key for Revision Booklets A and B

Use this document in two stages. First read the **help sheet** for the topic you are revising and try the questions in Booklet A or B on your own. Then check your work against the **worked solutions** at the back. If an answer is different, compare your working line by line to find where it went wrong.

Part 1 Help sheet: key ideas and formulas for every yearly exam topic

1 Financial mathematics · 2 Surface area and volume · 3 Algebraic techniques and indices · 4 Pythagoras and trigonometry · 5 Data analysis · 6 Equations · 7 Linear relationships

Part 2 Worked solutions: Booklet A

Part 3 Worked solutions: Booklet B

Part 1 · Help sheet

1 Financial mathematics

Ways of being paid

- **Wage:** paid per hour. Pay = hourly rate × hours.
- **Salary:** a fixed amount per year, paid in equal parts.
- **Commission:** a percentage of sales. Sometimes paid on top of a **retainer** (a fixed weekly amount).
- **Overtime:** time-and-a-half = rate × 1.5; double time = rate × 2.

Time conversions for pay
1 year = 52 weeks = 26 fortnights = 12 months
Weekly pay = salary ÷ 52
Fortnightly pay = salary ÷ 26
Monthly pay = salary ÷ 12

Example: \$25/h for 38 h, then 4 h at time-and-a-half.
Normal: 25 × 38 = \$950. Overtime: 4 × 25 × 1.5 = \$150. Total = **\$1100**.

Tax and deductions

- **Gross pay** = total earned before anything is taken out.
- **Net pay** = gross pay − deductions (tax, super, union fees …).
- **Taxable income** = gross income − allowable tax deductions (work-related costs).

Buying and selling

- Profit = selling price − cost price
- % profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$
- Discount = marked price × rate
- Sale price = marked price − discount (or marked price × (100% − rate))
- GST is 10%: price with GST = price × 1.1

Simple interest: $I = Prn$ P = principal (amount invested), r = interest rate per year as a decimal, n = number of years.
Months → years: divide by 12 (e.g. 30 months = 2.5 years). Total value = $P + I$.

Example: \$4000 at 3% p.a. for 18 months. $I = 4000 \times 0.03 \times 1.5 =$ **\$180**.

Watch out: change the % into a decimal (3.5% = 0.035) and the time into years before you use $I = Prn$.

2 Surface area and volume (including measurement)

Units
mm → cm → m → km: ×10, ×100, ×1000 (divide to go the other way)
1 kL = 1000 L 1 L = 1000 mL
1 cm³ = 1 mL 1 L = 1000 cm³
1 m³ = 1000 L
Area: 1 m² = 10 000 cm², 1 cm² = 100 mm²

Precision and error

- **Precision** = the smallest unit marked on the instrument.
- **Absolute error** = ½ × precision.
- A measurement of 15 cm (to the nearest cm) lies between 14.5 cm and 15.5 cm.
- % error = $\frac{\text{absolute error}}{\text{measurement}} \times 100\%$

Shape	Area / perimeter	Notes
Rectangle	$A = lb$	Perimeter = add up all the sides
Triangle	$A = \frac{1}{2}bh$	height is perpendicular to the base
Parallelogram	$A = bh$	use the perpendicular height, not the slanted side
Trapezium	$A = \frac{1}{2}h(a + b)$	a, b are the parallel sides
Kite and rhombus	$A = \frac{1}{2}xy$	x, y are the diagonals
Circle	$C = 2\pi r = \pi d$, $A = \pi r^2$	if given the diameter, halve it to get r
Composite shape	split into simple shapes, or big shape − cut-out	find any missing side lengths first

Solid	Volume	Surface area (closed)
Any prism	$V = Ah$ (area of cross-section × height)	add the areas of all faces (draw the net)
Rectangular prism	$V = lwh$	$SA = 2(lw + lh + wh)$
Cylinder	$V = \pi r^2 h$	$SA = 2\pi r^2 + 2\pi rh$ (two circles + curved surface)

Example: Cylinder $r = 3$ cm, $h = 10$ cm. $V = \pi \times 3^2 \times 10 = 282.7$ cm³ = 282.7 mL. $SA = 2\pi(3^2) + 2\pi(3)(10) = 56.5 + 188.5 = 245.0$ cm².

Watch out: area is in square units (cm²), volume in cubic units (cm³). An open container has one less face, so leave out the missing top.

3 Algebraic techniques and indices

Simplifying

- **Like terms** have the same pronumerals with the same powers: $3x$ and $-5x$ are like terms; $3x$ and $3x^2$ are not.
- Only add or subtract like terms: $7x + 2y - 3x = 4x + 2y$.
- Multiply: numbers × numbers, letters × letters: $6m \times 4n = 24mn$.
- Divide: write as a fraction and cancel: $20ab \div 5a = 4b$.
- Substitution: replace each letter with its value in brackets, then use BODMAS. If $q = 3$, $3q^2 = 3 \times 3^2 = 27$ (square *first*).

Expanding (removing brackets)

- Multiply every term inside by the term outside: $4(3x - 2) = 12x - 8$.
- A negative outside changes every sign: $-3(4k - 7) = -12k + 21$.
- Expand each bracket, then collect like terms.

Factorising (putting brackets back)

- Find the highest common factor (HCF) of the numbers *and* the letters, and put it outside.
- $12y^2 - 18y = 6y(2y - 3)$.
- Check by expanding your answer.

Index laws

$a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$

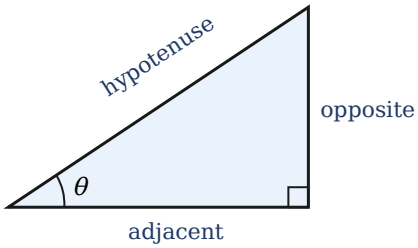
$a^0 = 1$ $a^{-n} = \frac{1}{a^n}$ $(ab)^n = a^n b^n$

Example: $2a^3b \times 6ab^2 = (2 \times 6) \times a^{3+1} \times b^{1+2} = \mathbf{12a^4b^3}$. $12y^9 \div 4y^3 = (12 \div 4)y^{9-3} = \mathbf{3y^6}$.

Watch out: multiply the coefficients (the numbers) but *add* the indices. A letter on its own has index 1 ($a = a^1$).

4 Pythagoras' theorem and trigonometry

Pythagoras' theorem (right-angled triangles only)
$c^2 = a^2 + b^2$, where c is the **hypotenuse**: the longest side, opposite the right angle.
Finding the hypotenuse: $c = \sqrt{(a^2 + b^2)}$ (**add** the squares)
Finding a shorter side: $a = \sqrt{(c^2 - b^2)}$ (**subtract** the squares)



Example: Hypotenuse 13, one side 5. Other side = $\sqrt{(13^2 - 5^2)} = \sqrt{144} = \mathbf{12}$.

Trigonometry: label the sides from the angle θ you are using.

SOH CAH TOA
$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$ $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$ $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$

Finding a side

1. Label opp, adj and hyp from the given angle.
2. Choose the ratio that uses the side you know and the side you want.
3. If the unknown is on top: multiply.
   $x = 20 \times \sin 35°$
4. If the unknown is on the bottom: divide.
   $x = 20 \div \cos 35°$

Finding an angle

- Work out the ratio, then use the inverse key (shift):
  $\cos\theta = \frac{1.3}{4.8} \rightarrow \theta = \cos^{-1}(1.3 \div 4.8) = 74°$
- Degrees and minutes: 1° = 60′. Use the ° ′ ″ key. 32.5° = 32°30′.
- Check your calculator is in **degrees** mode (D on the screen).

Angles of elevation and depression are measured from the **horizontal**: up to an object (elevation) or down to an object (depression).

5 Data analysis

Types of data

- **Categorical:** words or categories (colour, car type, S/M/L, poor/good/excellent).
- **Numerical discrete:** counted, so only whole values (number of pets, goals).
- **Numerical continuous:** measured, so any value in a range (mass, time, length, temperature).

Measures of centre and spread

- **Mean** = $\frac{\text{sum of scores}}{\text{number of scores}}$
- **Median** = middle score when the scores are in order. With an even number of scores, average the two middle ones. Its position is the $\frac{n+1}{2}$th score.
- **Mode** = the most common score.
- **Range** = highest − lowest.

Shape of a distribution

- **Symmetrical:** both sides look roughly the same.
- **Positively skewed:** scores bunch on the left (low end) with a long tail to the right.
- **Negatively skewed:** scores bunch on the right (high end) with a tail to the left.
- **Cluster:** where most scores are grouped.
  Outlier: a score far from the rest.

Outliers change the **mean** a lot but the **median** very little, so the median describes data with an outlier better.

Comparing two sets (back-to-back stem-and-leaf plots): the higher median (or mean) means "better" or "more"; the smaller range means more **consistent**.

Reading graphs: read stem-and-leaf plots with the key (3 | 5 = 35). In a back-to-back plot, the left-hand leaves are read from the stem outward. Histogram: total = add the column heights.

Example: 3, 5, 6, 6, 8, 20. Mean = 48 ÷ 6 = 8. Median = (6 + 6) ÷ 2 = 6. Mode = 6. Range = 20 − 3 = 17. The outlier 20 pulls the mean up, so the median is the better measure.

Percentage of a total: $\frac{\text{part}}{\text{total}} \times 100\%$. Check that your categories add up to the total.

6 Equations

- Whatever you do to one side, do the same to the other side.
- Undo operations in reverse order using inverse operations: + and − undo each other; × and ÷ undo each other.
- **Brackets:** expand first.
- **Pronumerals on both sides:** subtract the smaller pronumeral term from both sides first.
- **Fractions:** multiply both sides by the denominator.
- **Check:** substitute your answer back into the original equation.
- **Word problems:** pick a letter for the unknown, write the equation, solve it, and answer in words with units.

Example 1: $5x - 4 = 2x + 11$
$-2x$: $3x - 4 = 11$
$+4$: $3x = 15$
$\div 3$: $\boldsymbol{x = 5}$ Check: $21 = 21$ ✓

Example 2: $2(a + 3) = 18$
Expand: $2a + 6 = 18 \rightarrow 2a = 12 \rightarrow \boldsymbol{a = 6}$

Example 3: $\frac{x}{3} + 4 = 9 \rightarrow \frac{x}{3} = 5 \rightarrow \boldsymbol{x = 15}$

7 Linear relationships

Key formulas

Gradient $m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$

Gradient–intercept form: $y = mx + c$
m = gradient, c = y-intercept

Midpoint $= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Distance $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

- Positive gradient: line goes up from left to right. Negative gradient: line goes down.
- Horizontal line $y = k$: gradient 0. Vertical line $x = k$: gradient undefined.
- **Parallel lines** have the same gradient.

- **Graphing:** make a table of values (e.g. $x = -1$, 0, 1, 2), plot the points and rule a line through them.
- ***x*-intercept:** set $y = 0$ and solve. ***y*-intercept:** set $x = 0$.
- **Does a point lie on the line?** Substitute x and check you get the same y.
- **Real-life lines:** in $C = 90 + 65h$, the 90 is a fixed starting cost (the y-intercept) and 65 is the rate per hour (the gradient).

Example: Line through (1, 3) and (4, 9).
$m = \frac{9 - 3}{4 - 1} = 2$. Substitute (1, 3): $3 = 2(1) + c$, so $c = 1$.
Equation: $\boldsymbol{y = 2x + 1}$. Midpoint = (2.5, 6).

Worked solutions

Booklet A

Answers are shown in **green bold**. Rounded answers follow the rounding asked for in the question.

Section 1: Data

1 **a** **Numerical, discrete** (pets are counted) **b** **Categorical** (colours are words) **c** **Numerical, continuous** (mass is measured) **d** **Numerical, discrete** (messages are counted) **e** **Numerical, continuous** (time is measured) **f** **Categorical** (sizes are labels)

2 Dots: 0 → 4, 1 → 8, 2 → 6, 3 → 3, 4 → 2, 5 → 1, 9 → 1

a 4 + 8 + 6 + 3 + 2 + 1 + 1 = **25 students**

b **Positively skewed**: the scores bunch at the low end with a tail to the right.

c **Between 0 and 2 pets**

d **Yes, 9 pets** (far away from the other scores)

e **Mode = 1 pet** (tallest column of dots)

3 Frequencies: 1 → 3, 2 → 7, 3 → 5, 4 → 6, 5 → 2, 6 → 1

a 3 + 7 + 5 + 6 + 2 + 1 = **24 houses** **b** **2 people** (tallest column) **c** 6 − 1 = **5** **d** 4, 5 or 6 people: 6 + 2 + 1 = **9 houses**

4 **a** **41**

b Sum = 128, so mean = 128 ÷ 8 = **16**. Ordered: 9, 11, 12, 13, 13, 14, 15, 41 → median = (13 + 13) ÷ 2 = **13**

c Sum = 87, so mean = 87 ÷ 7 = **12.4** (1 d.p.). Ordered: 9, 11, 12, 13, 13, 14, 15 → median = **13**

d The **mean** changed more (16 → 12.4) while the median stayed at 13. The **median** better describes the data.

5 Scores: 23, 27, 31, 32, 35, 35, 38, 40, 42, 44, 46, 47 (12 scores)

a Sum = 440, so mean = 440 ÷ 12 = 36.666… = **36.7**

b 6th and 7th scores: (35 + 38) ÷ 2 = **36.5**

c **35** **d** 47 − 23 = **24**

6 Apple 12, Banana 9, Mango 15, Grapes 6, Orange 3

a **Mango** **b** 15 − 6 = **9 more students** **c** 12 + 9 + 15 + 6 + 3 = **45 students** **d** $\frac{12 + 9}{45} \times 100\% =$ **46.7%**

7 9X: 21, 25, 28, 30, 32, 33, 35, 36, 38, 41 | 9Y: 14, 18, 26, 29, 37, 40, 42, 45, 47, 49 (10 marks each)

a 9X: (32 + 33) ÷ 2 = **32.5** 9Y: (37 + 40) ÷ 2 = **38.5**

b 9X: 41 − 21 = **20** 9Y: 49 − 14 = **35**

c **9Y**, because it has the higher median (38.5 compared with 32.5).

d **9X**, because its range is smaller (20 compared with 35), so its marks are closer together.

Section 2: Surface area and volume

8 **a** 2.4 × 1000 = **2400 L** **b** 3750 ÷ 1000 = **3.75 L** **c** 0.6 × 1000 = **600 mL** **d** 8200 ÷ 1000 = **8.2 kL** **e** 4000 ÷ 1000 = **4 L** **f** 2.5 × 1000 = **2500 L**

9 Absolute error = ½ × precision

a (i) **1 mm** (ii) **0.5 mm** **b** (i) **25 mL** (ii) **12.5 mL**

c (i) **0.1 kg** (ii) **0.05 kg** **d** (i) **2°C** (ii) **1°C**

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10 **a** Parallelogram: $A = 14 \times 8 =$ **112 m²** (the 9 m slanted side is not used) **b** Kite: $A = ½ \times 12 \times 18 =$ **108 cm²**

c Trapezium: $A = ½ \times 7 \times (9 + 15) =$ **84 cm²** **d** Rhombus: $A = ½ \times 10 \times 16 =$ **80 mm²**

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11 **a** (i) $8 + 15 + 17 =$ **40 cm** (ii) $½ \times 15 \times 8 =$ **60 cm²**

b (i) $7 + 24 + 25 =$ **56 mm** (ii) $½ \times 7 \times 24 =$ **84 mm²** (use the two sides that meet at the right angle)

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12 **a** $r = 4.2$ cm. (i) $C = 2 \times \pi \times 4.2 = 26.389\ldots =$ **26.4 cm** (ii) $A = \pi \times 4.2^2 = 55.417\ldots =$ **55.4 cm²**

b Diameter 15 m, so $r = 7.5$ m. (i) $C = \pi \times 15 = 47.123\ldots =$ **47.1 m** (ii) $A = \pi \times 7.5^2 = 176.714\ldots =$ **176.7 m²**

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13 Missing sides: the right-hand edges are 5 m and 6 m ($5 + 6 = 11$ ✓); the step across is 6 m ($10 + 6 = 16$ ✓).

a $P = 10 + 5 + 6 + 6 + 16 + 11 =$ **54 m**

b Full rectangle − missing corner $= 16 \times 11 - 6 \times 5 = 176 - 30 =$ **146 m²**

c $146 \times \$38.50 =$ **\$5621.00**

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14 **a** $V = \pi \times 4^2 \times 11 = 176\pi = 552.920\ldots =$ **552.9 cm³**

b 1 cm³ = 1 mL, so the can holds **552.9 mL** (about 553 mL)

c $SA = 2\pi(4^2) + 2\pi(4)(11) = 32\pi + 88\pi = 120\pi = 376.991\ldots =$ **377.0 cm²**

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15 **a** $V = 32 \times 20 \times 12 =$ **7680 cm³**

b 7680 cm³ = 7680 mL = **7.68 L**

c $SA = 2(32 \times 20 + 32 \times 12 + 20 \times 12) = 2(640 + 384 + 240) = 2 \times 1264 =$ **2528 cm²**

Section 3: Mixed topics

16 **a** $2(6) + 3 =$ **15** **b** $6^2 - 3 = 36 - 3 =$ **33**

c $3 \times 3^2 = 3 \times 9 =$ **27** **d** $(6 - 3)^2 = 3^2 =$ **9**

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17 **a** x is the hypotenuse: $x^2 = 6.5^2 + 8.2^2 = 42.25 + 67.24 = 109.49$, so $x = \sqrt{109.49} = 10.46\ldots =$ **10.5 cm**

b 14 m is the hypotenuse: $y^2 = 14^2 - 9^2 = 196 - 81 = 115$, so $y = \sqrt{115} = 10.72\ldots =$ **10.7 m**

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18 **a** Discount = 20% × \$85 = \$17. Sale price = \$85 − \$17 = **\$68** (or 0.8 × 85)

b Saved = 15% × \$240 = **\$36**. Sale price = \$240 − \$36 = **\$204**

c Profit = \$799 − \$650 = **\$149**. % profit = $\frac{149}{650} \times 100\% = 22.92\ldots =$ **22.9%**

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19 **a** $(3 \times 5)x^{4+3} =$ **$15x^7$** **b** $(12 \div 4)y^{9-3} =$ **$3y^6$**

c $m^{2\times5} =$ **m^{10}** **d** $(2 \times 6)a^{3+1}b^{1+2} =$ **$12a^4b^3$**

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20 **a** $78\,260 \div 52 =$ **\$1505** **b** $78\,260 \div 26 =$ **\$3010** **c** $78\,260 \div 12 =$ **\$6521.67**

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21 **a** $(7x - 3x) + (2y + 5y) =$ **$4x + 7y$** **b** **$24mn$**

c $\frac{20ab}{5a} =$ **$4b$** **d** $(9k + 2k) + (-4 + 10) =$ **$11k + 6$**

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22 **a** **$12x - 8$** **b** **$-10a - 15$**

c $2x + 14 + 3x - 12 =$ **$5x + 2$** **d** $10y - 5 - 3y + 18 =$ **$7y + 13$**

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23 **a** HCF 4: **$4(2x + 5)$** **b** HCF 5: **$5(3m - 2)$**

c HCF a: **$a(a + 6)$** **d** HCF $6y$: **$6y(2y - 3)$**

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24 **a** $I = 5400 \times 0.035 \times 4 =$ **$756**

b 30 months = 2.5 years. $I = 12\,000 \times 0.052 \times 2.5 =$ **$1560**

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25 **a** $4x = 28 \rightarrow$ **$x = 7$** **b** $x \div 3 = 7 \rightarrow$ **$x = 21$**

c $4m - 7 = 25 \rightarrow 4m = 32 \rightarrow$ **$m = 8$** **d** $6y - 3 = 4y + 15 \rightarrow 2y = 18 \rightarrow$ **$y = 9$**

e $5k - 3k - 6 = k + 4 \rightarrow 2k - 6 = k + 4 \rightarrow$ **$k = 10$**

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26 **a** Commission = 1.8% × \$85 000 = \$1530. Total = \$650 + \$1530 = **\$2180**

b 6% × \$23 400 = 0.06 × 23 400 = **\$1404**

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27 **a** 38 × \$28.40 = **\$1079.20**

b 45 − 38 = 7 overtime hours. 7 × \$28.40 × 1.5 = 7 × \$42.60 = **\$298.20**

c \$1079.20 + \$298.20 = **\$1377.40**

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28 **a** $C = 90 + 65 \times 3 = 90 + 195 =$ **\$285**

b $90 + 65h = 415 \rightarrow 65h = 325 \rightarrow h = 5$. The job took **5 hours**.

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29 **a** $h^2 = 4.8^2 - 1.3^2 = 23.04 - 1.69 = 21.35$, so $h = \sqrt{21.35} = 4.6206\ldots =$ **4.62 m**

b Adjacent = 1.3, hypotenuse = 4.8, so use cos: $\cos\theta = \frac{1.3}{4.8} \rightarrow \theta = \cos^{-1}(0.2708\ldots) = 74.29\ldots° =$ **74°**

Worked solutions

Booklet B

Section 1: Data

1 **a** **Numerical, discrete** (counted) **b** **Categorical** (car types are names)
c **Numerical, continuous** (measured) **d** **Numerical, discrete** (counted)
e **Numerical, continuous** (measured) **f** **Categorical** (ratings are words)

2 Dots: 2 → 1, 3 → 2, 4 → 4, 5 → 5, 6 → 6, 7 → 4, 8 → 2, 9 → 1
a 1 + 2 + 4 + 5 + 6 + 4 + 2 + 1 = **25 students**
b **Symmetrical** (roughly the same shape on both sides of the centre)
c **Between 4 and 7**
d **No outlier** (there are no gaps; every score is close to the others)
e **Mode = 6**

3 Frequencies: 0 → 2, 1 → 5, 2 → 8, 3 → 6, 4 → 3, 5 → 1
a 2 + 5 + 8 + 6 + 3 + 1 = **25 students** **b** **2 books**
c 5 − 0 = **5** **d** 0 or 1 book: 2 + 5 = **7 students**

4 **a** **30**
b Sum = 59, so mean = 59 ÷ 8 = 7.375 ≈ **7.4 min**. Ordered: 2, 3, 4, 4, 5, 5, 6, 30 → median = (4 + 5) ÷ 2 = **4.5 min**
c Sum = 29, so mean = 29 ÷ 7 = **4.1 min** (1 d.p.). Ordered: 2, 3, 4, 4, 5, 5, 6 → median = **4 min**
d The **mean** changed more (7.4 → 4.1); the median only moved from 4.5 to 4. The **median** better describes the data.

5 Scores: 8, 12, 15, 17, 19, 21, 21, 24, 28, 30, 33 (11 scores)
a Sum = 228, so mean = 228 ÷ 11 = 20.72… = **20.7**
b Middle (6th) score = **21**
c **21** **d** 33 − 8 = **25**

6 Bus 18, Car 14, Walk 10, Bike 6, Train 12
a **Bike** **b** 18 − 14 = **4 more students**
c 18 + 14 + 10 + 6 + 12 = **60 students** **d** $\frac{10 + 6}{60} \times 100\% =$ **26.7%**

7 Jai: 12, 15, 18, 19, 21, 22, 22, 24, 25, 27, 29 | Ben: 5, 9, 14, 17, 20, 26, 28, 31, 33, 36, 38 (11 games each)
a Jai: 6th score = **22** Ben: 6th score = **26**
b Jai: 29 − 12 = **17** Ben: 38 − 5 = **33**
c **Ben**, because his median is higher (26 compared with 22).
d **Jai**, because his range is much smaller (17 compared with 33).

Section 2: Surface area and volume

8 **a** 5.3 × 1000 = **5300 L** **b** 1250 ÷ 1000 = **1.25 L** **c** 3.75 × 1000 = **3750 mL**
d 640 ÷ 1000 = **0.64 kL** **e** 7500 ÷ 1000 = **7.5 L** **f** 0.8 × 1000 = **800 L**

9 **a** (i) **1 cm** (ii) **0.5 cm** **b** (i) **50 mL** (ii) **25 mL**
c (i) **0.5 kg** (ii) **0.25 kg** **d** (i) **10 km/h** (ii) **5 km/h**

10 **a** Parallelogram: 18 × 9.5 = **171 cm²** (not the 11 cm side)

b Kite: ½ × 14 × 20 = **140 m²**

c Trapezium: ½ × 8 × (6 + 13) = **76 m²**

d Rhombus: ½ × 9 × 15 = **67.5 cm²**

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11 **a** (i) 20 + 21 + 29 = **70 m** (ii) ½ × 21 × 20 = **210 m²**

b (i) 12 + 16 + 20 = **48 cm** (ii) ½ × 16 × 12 = **96 cm²**

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12 **a** $r = 8.5$ mm. (i) $C = 2 \times \pi \times 8.5 = 53.407\ldots =$ **53.4 mm** (ii) $A = \pi \times 8.5^2 = 226.980\ldots =$ **227.0 mm²**

b Diameter 9 cm, so $r = 4.5$ cm. (i) $C = \pi \times 9 = 28.274\ldots =$ **28.3 cm** (ii) $A = \pi \times 4.5^2 = 63.617\ldots =$ **63.6 cm²**

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13 Check the sides: 6 + 4 = 10 ✓ and 5 + 7 = 12 ✓

a $P = 5 + 6 + 7 + 4 + 12 + 10 =$ **44 m** of skirting board

b Full rectangle − missing corner = 12 × 10 − 7 × 6 = 120 − 42 = **78 m²**

c 78 × \$45.90 = **\$3580.20**

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14 **a** $V = \pi \times 1.5^2 \times 2 = 4.5\pi = 14.137\ldots =$ **14.14 m³**

b 1 m³ = 1000 L: 14.137... × 1000 = 14 137.1... = **14 137 L** (use the unrounded volume)

c $SA = 2\pi(1.5^2) + 2\pi(1.5)(2) = 4.5\pi + 6\pi = 10.5\pi = 32.986\ldots =$ **33.0 m²**

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15 **a** $V = 45 \times 25 \times 16 =$ **18 000 cm³**

b 18 000 cm³ = 18 000 mL = **18 L**

c $SA = 2(45 \times 25 + 45 \times 16 + 25 \times 16) = 2(1125 + 720 + 400) = 2 \times 2245 =$ **4490 cm²**

Section 3: Mixed topics

16 **a** 4(5) − 2 = **18**

b $5^2 + 3(2) = 25 + 6 =$ **31**

c $2 \times 2^2 = 2 \times 4 =$ **8**

d $(5 + 2)^2 = 7^2 =$ **49**

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17 **a** x is the hypotenuse: $x^2 = 4.8^2 + 11.5^2 = 23.04 + 132.25 = 155.29$, so $x = \sqrt{155.29} = 12.46\ldots =$ **12.5 m**

b 17.6 cm is the hypotenuse: $y^2 = 17.6^2 - 10.3^2 = 309.76 - 106.09 = 203.67$, so $y = \sqrt{203.67} = 14.27\ldots =$ **14.3 cm**

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18 **a** Discount = 25% × \$120 = \$30. Sale price = **\$90**

b Saved = 12% × \$360 = **\$43.20**. Sale price = \$360 − \$43.20 = **\$316.80**

c Profit = \$378 − \$280 = **\$98**. % profit = $\frac{98}{280} \times 100\% =$ **35%**

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19 **a** $(4 \times 7)k^{5+2} =$ **$28k^7$**

b $(18 \div 3)w^{8-2} =$ **$6w^6$**

c $p^{3\times4} =$ **p^{12}**

d $(5 \times 3)m^{2+4}n^{3+1} =$ **$15m^6n^4$**

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20 **a** 92 040 ÷ 52 = **\$1770**

b 92 040 ÷ 26 = **\$3540**

c 92 040 ÷ 12 = **\$7670**

---

21 **a** $(8p + 2p) + (-3q + 7q) =$ **$10p + 4q$**

b **$15ac$**

c $\frac{24xy}{8y} =$ **$3x$**

d $(4t - t) + (9 - 3) =$ **$3t + 6$**

---

22 **a** **$12m + 30$**

b **$-12k + 21$**

c $3a + 12 + 2a - 10 =$ **$5a + 2$**

d $12p - 8 - 2p - 6 =$ **$10p - 14$**

---

23 **a** HCF 3: **$3(2k + 7)$**

b HCF 7: **$7(2n - 5)$**

c HCF p: **$p(p - 9)$**

d HCF $5x$: **$5x(2x + 3)$**

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24 **a** $I = 8600 \times 0.045 \times 3 =$ **$1161**

b 18 months = 1.5 years. $I = 15\,000 \times 0.038 \times 1.5 =$ **$855**

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25 **a** $7x = 49 \rightarrow$ ***x* = 7**

b $x \div 4 = 5 \rightarrow$ ***x* = 20**

c $4a + 4 = 36 \rightarrow 4a = 32 \rightarrow$ ***a* = 8**

d $6p + 8 = 4p + 20 \rightarrow 2p = 12 \rightarrow$ ***p* = 6**

e $8n - 2n - 10 = 3n + 11 \rightarrow 6n - 10 = 3n + 11 \rightarrow 3n = 21 \rightarrow$ ***n* = 7**

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26 **a** Commission = 3.5% × \$36 200 = \$1267. Total = \$480 + \$1267 = **$1747**

b 5% × \$18 900 = 0.05 × 18 900 = **$945**

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27 **a** 38 × \$24.60 = **$934.80**

b 41 − 38 = 3 overtime hours. 3 × \$24.60 × 1.5 = 3 × \$36.90 = **$110.70**

c \$934.80 + \$110.70 = **$1045.50**

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28 **a** $C = 4.5 + 2.3 \times 12 = 4.5 + 27.6 =$ **$32.10**

b $4.5 + 2.3d = 50.5 \rightarrow 2.3d = 46 \rightarrow d = 20$. The taxi travelled **20 km**.

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29 **a** $d^2 = 25^2 - 18^2 = 625 - 324 = 301$, so $d = \sqrt{301} = 17.349\ldots =$ **17.35 m**

b Opposite = 18, hypotenuse = 25, so use sin: $\sin\theta = \frac{18}{25} \rightarrow \theta = \sin^{-1}(0.72) = 46.05\ldots° =$ **46°**

Small differences in the last decimal place can come from rounding too early. Keep full calculator values until the final step.